Oxidation of one molecule of NADH + H^+ gives rise to 3 molecules of ATP and oxidation of one molecule $FADH_2$ produces 2 molecules of ATP within a mitochondrion. But cytoplasmic NADH + H^+ yields only two ATPs through external NADH dehydrogenase. Therefore, two reduced coenzyme (NADH + H^+) molecules from glycolysis being extra mitochondrial will yield 2 × 2 = 4 ATP molecules instead of 6 ATPs (Figure 14.10). The Mechanism of mitochondrial ATP synthesis is based on Chemiosmotic hypothesis. According to this theory electron carriers present in the inner mitochondrial membrane allow for the transfer of protons (H^+). For the production of single ATP, 3 protons (H^+) are needed. The terminal oxidation of external NADH bypasses the first phosphorylation site and hence only two ATP molecules are produced per external NADH oxidised through mitochondrial electron transport chain. However, in those animal tissues in which malate shuttle mechanism is present, the oxidation of external NADH will yield almost 3 ATP molecules.

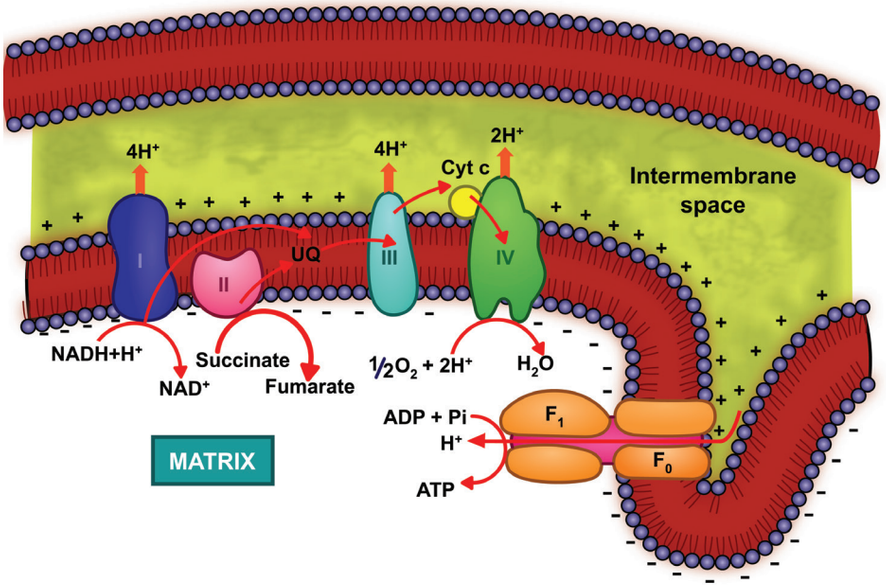


Figure 14.10: Electron Transport Chain and Terminal Oxidation

Complete oxidation of a glucose molecule in aerobic respiration results in the net gain of **36 ATP molecules in plants** as shown in table 14.2. Since huge amount of energy is generated in mitochondria in the form of ATP molecules they are called ***'power house of the cell'.*** In the case of aerobic prokaryotes due to lack of mitochondria each molecule of glucose produces 38 ATP molecules.

DO YOU KNOW?

Abnormal rise in respiratory rate of ripening in fruits is called **Climacteric**. Examples are apple, banana, mango, papaya, pear.

Recent view

When the cost of transport of ATPs from matrix into the cytosol is considered, the number will be **2.5 ATPs for each NADH + H^+ and 1.5 ATPs for each $FADH_2$** oxidised during electron transport system.

Table 14.2: Net Products gained during aerobic respiration per glucose molecule.

Stages	CO_2	ATP	Reduced NAD^+	Reduced FAD	Total ATP Production
Glycolysis	0	2	2 (2 × 2 = 4)	0	6
Link reaction	2	0	2 (2 × 3 = 6)	0	6
Krebs cycle	4	2	6 (6 × 3 = 18)	2 (2 × 2 = 4)	24
Total	**6**	**4 ATPs**	**28 ATPs**	**4 ATPs**	**36 ATPs**

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